

Building a SIEM Dashboard Using Splunk

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1. Introduction

A Security Information and Event Management (SIEM) system provides centralized collection, analysis, and visualization of security events. It helps security analysts monitor activity from different systems and identify potentially suspicious behavior.

This project focuses on **building a SIEM dashboard using Splunk**. A virtualized lab environment was created using Kali Linux, Ubuntu Linux, and Windows 11. Ubuntu was used as the log-generating server, while Splunk Enterprise was deployed on Windows 11 for centralized security monitoring.

The Splunk Universal Forwarder installed on Ubuntu collected authentication, Apache web-server, and kernel/firewall logs and forwarded them to Splunk Enterprise. SPL queries were then used to analyze the collected data and create visualizations for security monitoring.

The final dashboard provides visibility into **SSH authentication activity, Apache HTTP activity, firewall-blocked traffic, source IP addresses, and overall security events**.

2. Project Objective

The main objective of this project is to develop a centralized **SIEM dashboard in Splunk** capable of collecting, analyzing, and visualizing security events generated within a controlled virtual lab environment.

The project aims to:

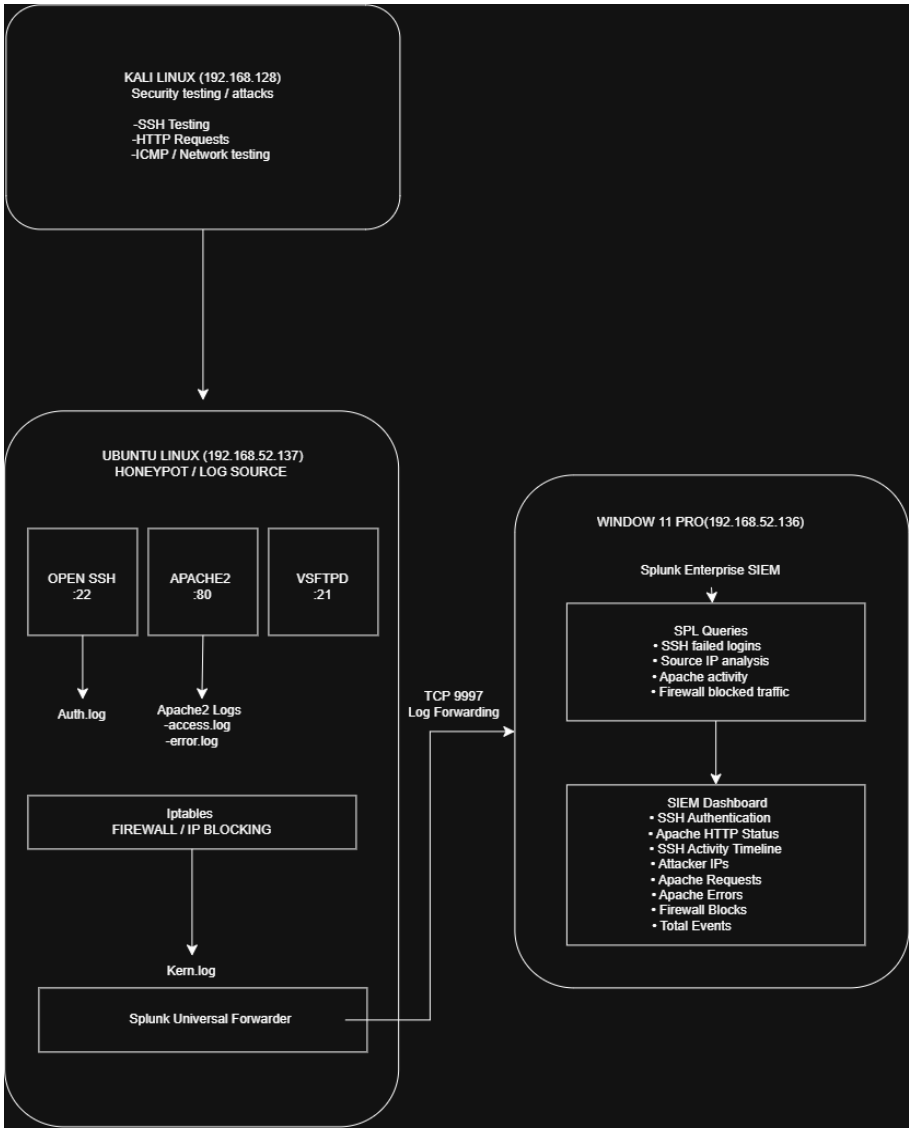
- Build a functional SIEM monitoring environment.
- Collect logs from an Ubuntu Linux system.
- Forward logs to Splunk Enterprise.
- Create SPL queries for security analysis.
- Identify failed SSH authentication and suspicious activity.
- Monitor Apache web-server activity.
- Monitor firewall-blocked traffic.
- Create an interactive security dashboard.
- Demonstrate centralized security monitoring.

3. Lab Environment

System	IP Address	Purpose
Kali Linux	192 . 168 . 52 . 128	Security testing
Ubuntu Linux	192 . 168 . 52 . 137	Log-generating server
Windows 11 Pro	192 . 168 . 52 . 136	Splunk Enterprise / SIEM

The environment was implemented using virtual machines, allowing security testing and log generation to be performed in an isolated laboratory environment.

4. Architecture



5. Tools & Technologies

Tool / Technology	Purpose
Kali Linux	Security testing and generating events
Ubuntu Linux	Log-generating server
Windows 11 Pro	Hosts Splunk Enterprise
OpenSSH	SSH authentication monitoring
Apache2	Web activity and HTTP logs
vsftpd	FTP service
iptables	Firewall and IP blocking
Splunk Universal Forwarder	Collects and forwards logs
Splunk Enterprise	SIEM and dashboard platform
SPL	Searching and analyzing security events
VMware	Virtualized lab environment

6. Splunk Configuration

Splunk Enterprise was installed on the Windows 11 Pro VM and configured as the centralized SIEM.

The Ubuntu system uses the **Splunk Universal Forwarder** to send security logs to Splunk Enterprise over **TCP port 9997**.

The logs are stored in the Splunk index:

ubuntulinux

6.1 Configured Log Sources

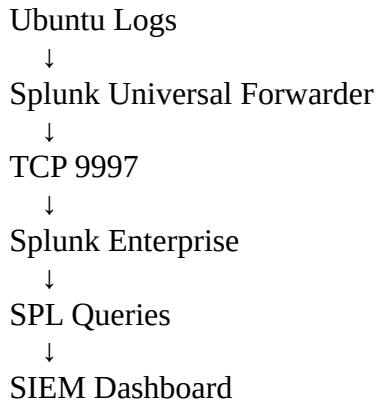
Ubuntu Log	Splunk Sourcetype
/var/log/auth.log	linux_secure
/var/log/apache2/access.log	apache:access
/var/log/apache2/error.log	apache:error
/var/log/kern.log	linux_kernel

```
root@osboxes: /home/osboxes/Downloads

/opt/splunkforwarder/var/log/splunk/btool.log
/opt/splunkforwarder/var/log/splunk/conf.log
/opt/splunkforwarder/var/log/splunk/first_install.log
/opt/splunkforwarder/var/log/splunk/health.log
/opt/splunkforwarder/var/log/splunk/license_usage.log
/opt/splunkforwarder/var/log/splunk/mergebuckets.log
/opt/splunkforwarder/var/log/splunk/mongod.log
/opt/splunkforwarder/var/log/splunk/mongod_upgrade.log
/opt/splunkforwarder/var/log/splunk/remote_searches.log
/opt/splunkforwarder/var/log/splunk/scheduler.log
/opt/splunkforwarder/var/log/splunk/search_messages.log
/opt/splunkforwarder/var/log/splunk/searchhistory.log
/opt/splunkforwarder/var/log/splunk/splunkd-utility.log
/opt/splunkforwarder/var/log/splunk/splunkd_access.log
/opt/splunkforwarder/var/log/splunk/splunkd_ui_access.log
/opt/splunkforwarder/var/log/splunk/wlm_monitor.log
$SPLUNK_HOME/var/log/splunk/configuration_change.log
/opt/splunkforwarder/var/log/splunk/configuration_change.log
$SPLUNK_HOME/var/log/splunk/license_usage_summary.log*
/opt/splunkforwarder/var/log/splunk/license_usage_summary.log
$SPLUNK_HOME/var/log/splunk/metrics.log
/opt/splunkforwarder/var/log/splunk/metrics.log
$SPLUNK_HOME/var/log/splunk/splunk_instrumentation_cloud.log*
/opt/splunkforwarder/var/log/splunk/splunk_instrumentation_cloud.log
$SPLUNK_HOME/var/log/splunk/splunkd.log
/opt/splunkforwarder/var/log/splunk/splunkd.log
$SPLUNK_HOME/var/log/watchdog/watchdog.log*
/opt/splunkforwarder/var/log/watchdog/watchdog.log
$SPLUNK_HOME/var/run/splunk/search_telemetry/*search_telemetry.json
$SPLUNK_HOME/var/spool/splunk/tracker.log*
/opt/splunkforwarder/var/spool/splunk/tracker.log

monitored Files:
$SPLUNK_HOME/etc/splunk.version
/var/log/apache2/access.log
/var/log/apache2/error.log
/var/log/auth.log
root@osboxes: /home/osboxes/Downloads#
```

6.2 Data Flow



7. Log Collection & Forwarding

The **Splunk Universal Forwarder** was installed on the Ubuntu server to collect security and system logs and forward them to Splunk Enterprise.

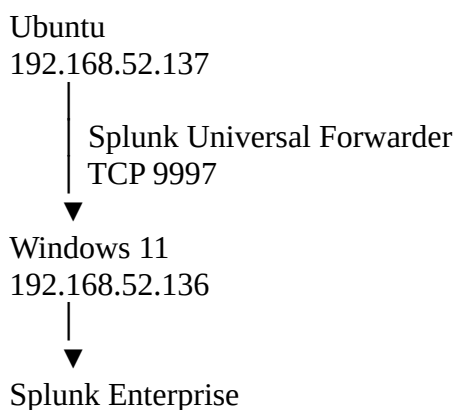
7.1 Monitored Logs

The following logs were configured:

- `/var/log/auth.log`
- `/var/log/apache2/access.log`
- `/var/log/apache2/error.log`
- `/var/log/kern.log`

7.2 Forwarding Configuration

The collected logs were sent to the Splunk Enterprise server on Windows 11 using:



7.3 Verification

The following SPL query was used to verify log ingestion:

```
index=ubuntulinux  
| stats count by sourcetype
```

The four configured sourcetypes were successfully received:

```
apache:access
apache:error
linux_kernel
linux_secure
```

This confirms successful **log collection, forwarding, and indexing** in Splunk.

8. Data Sources

The collected data provides visibility into:

- **SSH authentication** — successful and failed login activity
- **Apache access logs** — HTTP requests and responses
- **Apache error logs** — web-server errors
- **Kernel/firewall logs** — blocked network traffic

These sources form the foundation of the SIEM dashboard and allow security events to be analyzed from a centralized location.

9. SPL Queries & Security Analysis

Splunk Search Processing Language (SPL) was used to analyze the collected security logs and identify important events.

9.1 SSH Failed Login Analysis

```
index=ubuntulinux sourcetype=linux_secure "Failed password"
| rex "from\s+(?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| stats count by src_ip
| where count >= 3
| sort - count
```

This query extracts the source IP and identifies IPs generating repeated failed SSH authentication attempts.

9.2 Apache HTTP Status Analysis

```
index=ubuntulinux sourcetype=apache:access
| rex "\"\s(?<http_status>\d{3})\s"
| stats count by http_status
```

This identifies HTTP response codes such as **200** and **404**.

9.3 SSH Activity Over Time

```
index=ubuntulinux sourcetype=linux_secure ("Failed password" OR "Invalid user")
| timechart span=5m count
```

This visualizes SSH authentication activity over time.

9.4 Firewall Block Analysis

```
index=ubuntulinux sourcetype=linux_kernel "IPTABLES-DROP"
| rex "SRC=(?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| stats count as blocked_packets by src_ip
| sort - blocked_packets
```

This identifies source IPs generating blocked firewall traffic.

10. SIEM Dashboard

The analyzed data was converted into visual dashboard panels in Splunk. The dashboard provides a centralized view of authentication, web-server, and firewall activity.

11. Dashboard Panels

The Splunk SIEM dashboard was created to provide centralized visibility into different types of security activity generated in the lab environment.

11.1 SSH Failed Login Attempts

Displays repeated failed SSH authentication attempts and their source IP addresses.

11.2 Apache HTTP Status Codes

Shows HTTP response codes such as 200 and 404 generated by the Apache server.

11.3 SSH Authentication Activity Over Time

Displays SSH authentication activity using a time-based visualization.

11.4 Top SSH Attacker IPs

Shows the source IP addresses generating the highest number of failed SSH attempts.

11.5 Apache Web Requests Over Time

Visualizes the volume of Apache HTTP requests over time.

11.6 Apache Error Activity

Displays Apache server errors over time.

11.7 Firewall Blocked Traffic

Shows source IPs associated with IPTABLES - DROP events.

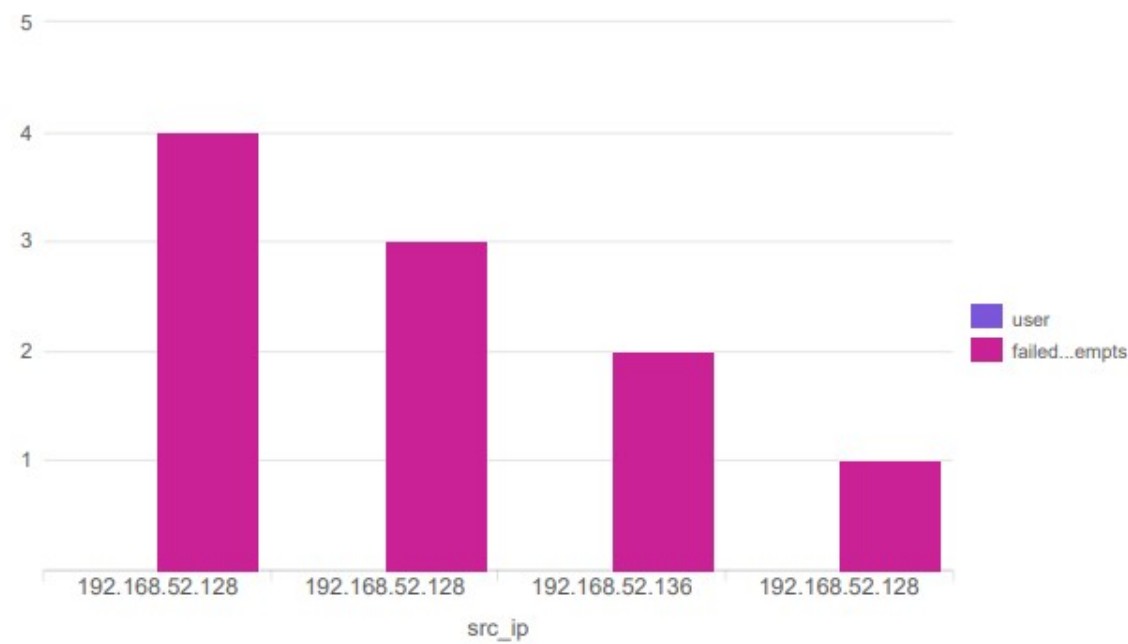
11.8 Total Security Events

Provides an overall count of events available in the `ubuntulinux` index.

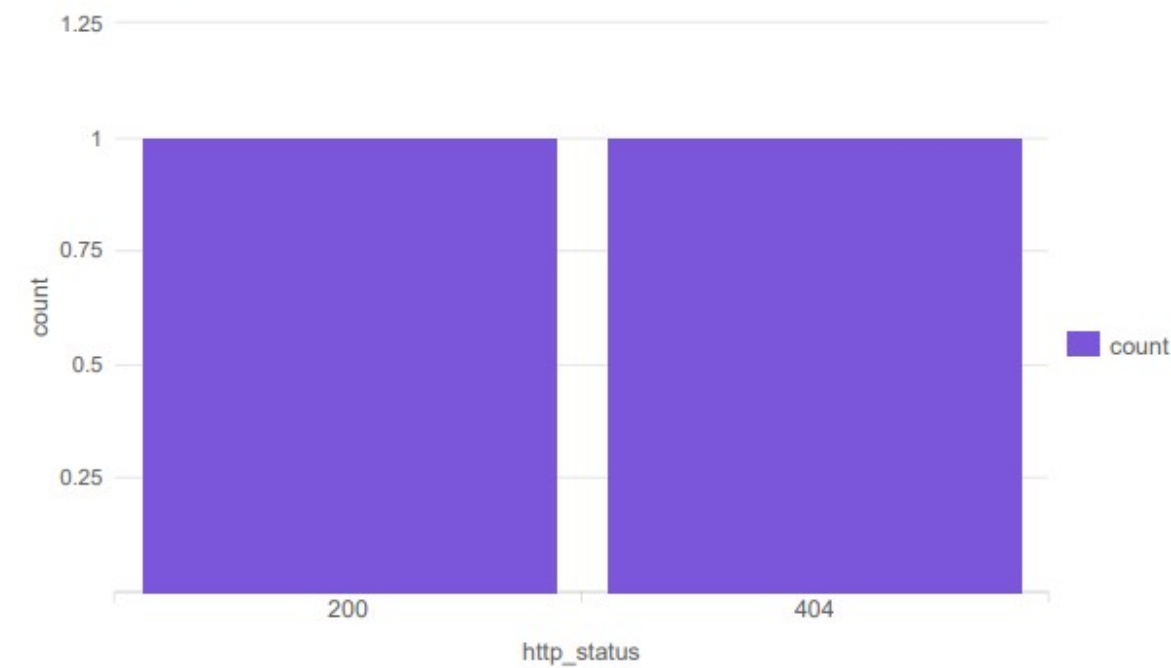
The completed dashboard combines authentication, web-server, and firewall information into a single monitoring interface.

Dashboard Screenshot

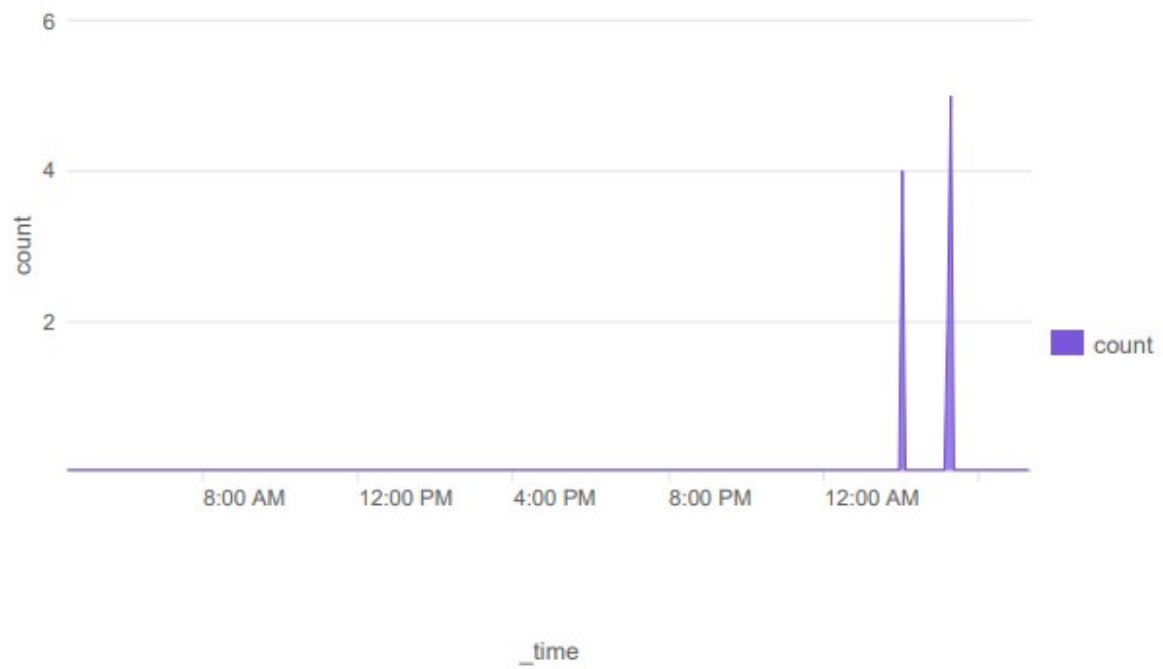
ssh failed login attempts



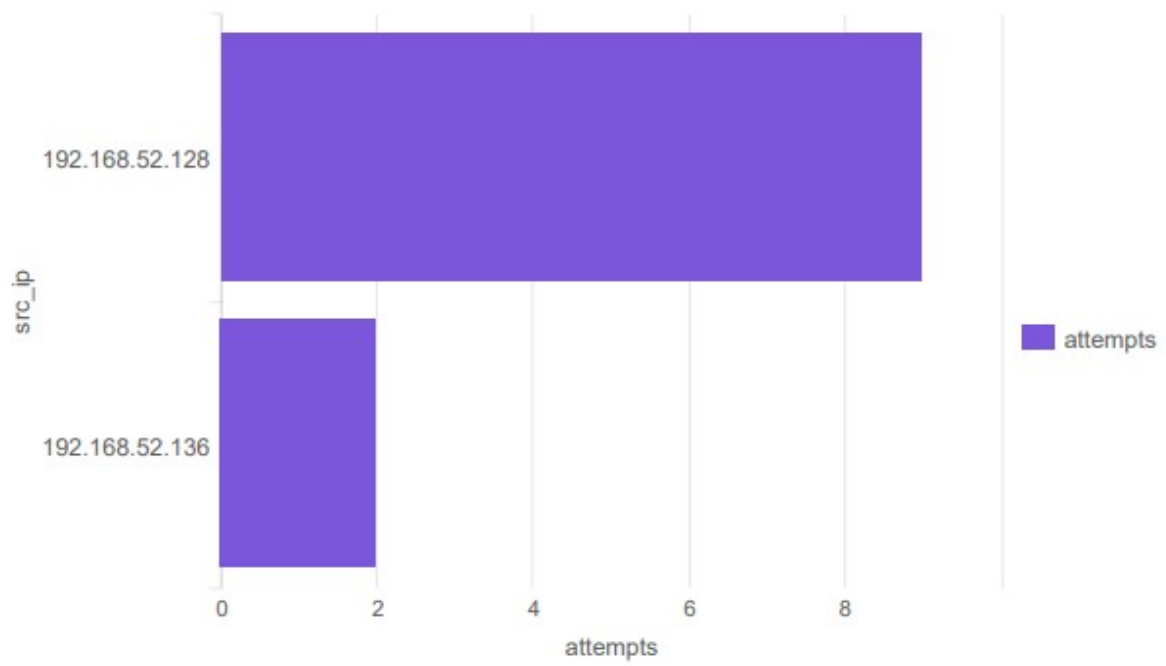
apache http status code

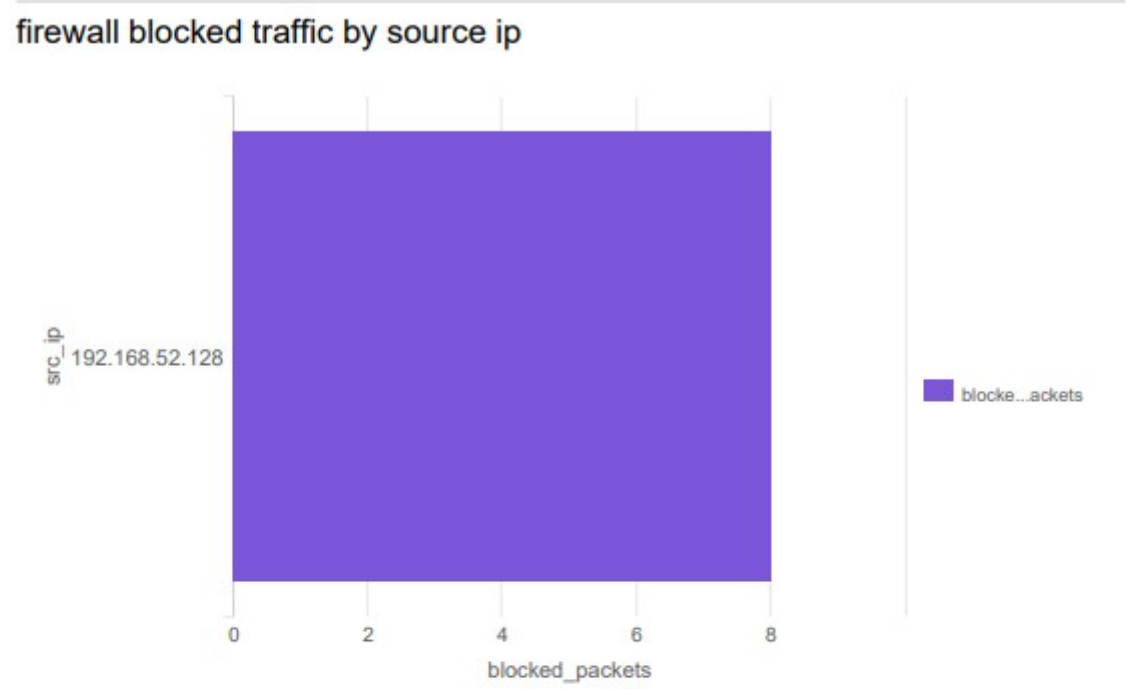


ssh authentication activity over time



top ssh attackers ip





total security events

2,011

_time

12. Testing & Results

The dashboard was tested using controlled activity from the Kali Linux environment. SSH failures, Apache requests/errors, and blocked firewall traffic were successfully recorded and displayed in Splunk.

13. Challenges & Solutions

During the implementation of the SIEM dashboard, several practical challenges were encountered.

Challenge 1 — Log Field Extraction

Some fields, such as the source IP and HTTP status, were not automatically extracted by Splunk.

Solution:

SPL `rex` commands were used to extract the required values directly from raw log events.

Challenge 2 — Firewall Log Visibility

The `iptables` DROP rules blocked traffic but did not automatically provide the required visibility in Splunk.

Solution:

A logging rule was added before the DROP rule, and `/var/log/kern.log` was forwarded to Splunk.

Challenge 3 — Multiple Log Sources

Security information was distributed across authentication, Apache, and kernel logs.

Solution:

The Splunk Universal Forwarder was configured to collect all four log sources and send them to a centralized Splunk index.

Challenge 4 — Dashboard Development

Raw security events were difficult to analyze directly.

Solution:

SPL queries and visualizations were combined into a centralized SIEM dashboard for easier monitoring and analysis.

14. Conclusion

The project successfully demonstrated the development of a **SIEM dashboard using Splunk**. Logs generated in the Ubuntu lab environment were collected using the Splunk Universal Forwarder, forwarded to Splunk Enterprise, analyzed using SPL, and presented through security-focused dashboard panels.

The final implementation provides centralized visibility into **SSH authentication, Apache activity, and firewall events**, demonstrating a practical SIEM monitoring workflow.

15. Future Scope

The SIEM dashboard can be further improved by:

- Adding more security data sources.
- Creating automated Splunk alerts for critical events.
- Adding advanced correlation and detection rules.
- Integrating automated incident-response actions.
- Adding more interactive dashboard filters.
- Extending the lab with additional attack scenarios.

16. Final Project Summary

The project successfully implemented a **Splunk-based SIEM dashboard** in a controlled virtual lab.

The complete workflow was:

Security Testing → Ubuntu Log Generation → Splunk Universal Forwarder → TCP 9997 → Splunk Enterprise → SPL Analysis → SIEM Dashboard

The dashboard provides centralized visibility into:

- SSH authentication activity
- Apache web activity
- Firewall-blocked traffic
- Source IP activity
- Security events over time

The project demonstrates practical skills in **Splunk, SPL, Linux log analysis, SIEM monitoring, and cybersecurity event visualization**.